

Continuous-Time Latent Dynamics for JEPA World Models: ODE Prediction in LeWorldModel

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Abstract. Joint Embedding Predictive Architecture (JEPA) world models such as LeWorldModel (LeWM) learn compact latent dynamics for model-based control, typically with a discrete autoregressive (AR) next-embedding predictor. We replace that predictor with an action-conditioned neural ODE $dz/d\tau = v_\theta(z, a)$, where one planning step is a fixed-step RK4 integration from $\tau=0$ to $\tau=1$. Encoder, SIGReg anti-collapse regularizer, projectors, and the CEM planning loop are unchanged. On the official PushT expert dataset with a fixed 1000-window subset (seed 3072; train 901 / val 99), both the discrete AR baseline and our ODE predictor achieve the same CEM success of **5/93** ($\approx 5.38\%$) on the val-tied protocol, while the ODE model attains lower train and validation losses (0.234 / 0.506 vs. 0.248 / 0.536). We report honest limitations: the 1k subset is far below the paper-scale regime that reports $\approx 96\%$ PushT success, and CEM is tied despite the loss improvement. All data are from the official Hugging Face release; no synthetic episodes are used.

1 Introduction

World models that plan in a compact latent space enable fast model-predictive control without decoding pixels at every step [6, 7]. Joint Embedding Predictive Architectures (JEPAs) [1] push this idea further by predicting future *embeddings* rather than reconstructions, and LeWorldModel (LeWM) [9] shows that a stable end-to-end JEPA can be trained from pixels with only two loss terms: next-embedding MSE and SIGReg isotropic-Gaussian regularization.

In LeWM, the predictor is a discrete autoregressive transformer block: given context latents z_t and action embeddings a_t , it directly outputs $\hat{z}_{t+1}$. Physical dynamics, however, are continuous in time. Recent continuous-time world models such as ODEWorld [8] argue that latent velocity fields $dz/dt = v_\theta(\cdot)$ better capture underlying flow and support flexible temporal resolution. This raises a concrete question for planning JEPA models:

Can we replace LeWM’s discrete AR predictor with an ODE integrate-to-next-step rule, keep the rest of the pipeline fixed, and retain competitive CEM planning on official PushT data?

Contributions. (1) We instantiate an **ODE predictor** inside the official LeWM JEPA: $dz/d\tau = v_\theta(z, a)$, with one-step prediction defined as RK4 integration over $\tau \in [0, 1]$ (fixed $N=4$ steps), actions held constant on the interval, and a zero-initialized linear velocity head for residual stability. (2) We evaluate on the **official** PushT expert HDF5 with a reproducible **1000-window** split (seed 3072;

901/99), matching encoder, SIGReg ($\lambda=0.09$, $M=1024$), optimizer, and CEM protocol between discrete and ODE runs ($\approx 18.07M$ parameters). (3) Empirically, ODE lowers train/val loss relative to discrete AR, while CEM success on the val-99 protocol is **tied** at 5/93 (5.38%). We explicitly contrast this with the paper-reported $96 \pm 2.83\%$ PushT figure under full-data training, and treat the 1k regime as a controlled ablation rather than a claim of paper-scale performance.

2 Related Work

JEPA world models and LeWM. JEPA methods learn predictive representations without pixel decoding [1, 3]. Collapse is a central failure mode; LeWM [9] stabilizes end-to-end pixel JEPA training with SIGReg [2], pairing MSE next-embedding prediction with a sketched isotropic-Gaussian regularizer. We adopt LeWM’s official encoder, SIGReg recipe, and CEM planner, changing only the predictor dynamics.

Continuous-time / ODE world models. Neural ODEs [4] parameterize continuous transformations via $dz/dt = v_\theta(z, t)$. ODE-World [8] introduces Physical-Time Flow: a continuous latent velocity field integrated by an ODE solver for video and control, emphasizing dynamical representation structure beyond discrete next-step prediction. Our setting is narrower and deliberately comparable: we keep LeWM’s JEPA loss and planning loop, and redefine a single discrete planning step as integrate $\tau:0 \rightarrow 1$ under an action-conditioned velocity field.

Latent MPC and CEM. Cross-entropy method (CEM) planning in latent world models [5, 7, 9] samples action sequences, rolls out the predictor, and ranks by a goal-conditioned cost. We use the official LeWM CEM evaluator on a fixed validation window set so discrete and ODE predictors share the same starts and feasibility filter.

3 Method

We keep the LeWM JEPA backbone [9] and replace only the discrete AR predictor with an ODE velocity field. Notation follows the official implementation (history length $H=3$, embedding dimension $D=192$).

3.1 Encoder and action embedding

Let $x_{t-H+1:t}$ be a stack of RGB observations (resolution 224×224). A ViT-tiny encoder f_ϕ (patch size 14, trained from scratch) maps

each frame to a CLS token; an MLP projector p yields latents

$$z_t = p(f_\phi(x_t)) \in \mathbb{R}^D. \quad (1)$$

Actions u_t are mapped by an embedder e_ψ to $a_t = e_\psi(u_t) \in \mathbb{R}^D$. Context tensors for one prediction step are $Z = (z_{t-H+1}, \dots, z_t)$ and $A = (a_{t-H+1}, \dots, a_t)$.

3.2 Discrete baseline predictor

The official LeWM predictor is an action-conditioned transformer (ConditionalBlock / AdaLN-zero) of depth 6, 16 heads, width D :

$$\hat{Z}_{t+1} = g_\theta^{\text{AR}}(Z, A), \quad (2)$$

interpreted as the next latent window (same shape as Z). A prediction head MLP q is applied after the predictor for the MSE term (unchanged in our ODE variant).

3.3 ODE predictor

We reinterpret the same transformer capacity as a **velocity field** v_θ , with a zero-initialized linear head W_v (adds $D^2 + D = 37,056$ parameters):

$$v_\theta(z, a) = W_v \text{TF}_\theta(z + \text{PE}, a), \quad (3)$$

where PE is a learnable positional embedding over the history axis and actions a condition AdaLN as in the AR baseline. Latent dynamics on a normalized step interval $\tau \in [0, 1]$ are

$$\frac{dz}{d\tau} = v_\theta(z(\tau), a), \quad z(0) = Z, \quad (4)$$

with a held fixed over the interval (piecewise-constant action, matching one CEM planning step). **One-step prediction** is the solution at $\tau=1$:

$$\hat{Z}_{t+1} = z(1) = Z + \int_0^1 v_\theta(z(\tau), a) d\tau. \quad (5)$$

We approximate the integral with fixed-step classical RK4 using $N=4$ substeps ($h=1/N$):

$$\begin{aligned} k_1 &= v_\theta(z, a), \\ k_2 &= v_\theta(z + \frac{h}{2}k_1, a), \\ k_3 &= v_\theta(z + \frac{h}{2}k_2, a), \\ k_4 &= v_\theta(z + h k_3, a), \\ z &\leftarrow z + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \quad (6)$$

Zero-init of W_v yields $v \approx 0$ at initialization, hence $z(1) \approx z(0)$ (stable residual ODE).

3.4 Training objective (SIGReg + prediction)

Following LeWM, the total loss is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda \mathcal{L}_{\text{SIGReg}}, \quad \lambda = 0.09, \quad (7)$$

where $\mathcal{L}_{\text{pred}}$ is MSE between projected predictions and target embeddings (stop-grad on targets), and $\mathcal{L}_{\text{SIGReg}}$ is the sketched Epps–Pulley statistic over $M=1024$ random projections and 17 quadrature knots [2, 9]. Only the predictor internals differ between discrete and ODE runs; encoder, projectors, SIGReg, and optimizer are shared.

Item	Experimental setup (PushT 1k subset).
Dataset	Official HF PushT expert HDF5
Subset / seed	1000 windows / 3072
Train / val	901 / 99
CEM eval set	val-99 $\rightarrow$ 93 feasible
Encoder	ViT-tiny, patch 14, 224 ² , scratch
SIGReg	$\lambda=0.09$, $M=1024$, knots 17
ODE solver	RK4, $N=4$, $\tau:0 \rightarrow 1$
Params (ODE)	18,071,534 ($\approx 18.07\text{M}$)
Train	100 epochs, AdamW 5×10^{-5}
Effective batch	128 ($64 \times \text{accum } 2$)

Table 2. PushT 1k $\times$ 100ep: loss and CEM (same split). Paper PushT success is under full-data training, shown for context only.

Method	Params	fit/loss	val/loss	CEM val-99
Discrete AR (LeWM)	18.03M	0.248	0.536	5/93 (5.38%)
ODE RK4 0 $\rightarrow$ 1 (ours)	18.07M	0.234	0.506	5/93 (5.38%)
LeWM paper (full data)	$\sim 15\text{M}$	—	—	$96 \pm 2.83\%^\dagger$

† Reported PushT success; not comparable to our 1k protocol.

3.5 Planning with CEM

At evaluation, Cross-Entropy Method (CEM) samples action sequences, rolls out the predictor in latent space from encoded pixels, and selects the plan minimizing distance to a goal embedding (official PushT CEM config). For the ODE model, each rollout step applies (5)–(6) instead of (2).

4 Experiments

4.1 Setup

Data. We use only the official Hugging Face PushT expert release (quentinll/lewm-pusht, HDF5). A reproducible subset of **1000** windows is drawn with seed **3072** and split by `spt.random_split([0.9, 0.1])` into **901** train / **99** validation windows (indices frozen; never regenerated between runs). Of the 99 val windows, **93** remain feasible under `goal_offset=25` for CEM (6 dropped).

Models and compute. Both predictors use ViT-tiny, history 3, $D=192$, SIGReg $\lambda=0.09$, AdamW lr 5×10^{-5} , weight decay 10^{-3} , grad clip 1.0, bf16, 100 epochs, frameskip 5. Hardware: RTX 4060 Laptop (8 GiB); batch 64 with gradient accumulation 2 (effective 128); `num_workers=0`. Parameter counts: discrete AR **18,034,478**; ODE **18,071,534** (+37,056 from v -head).

4.2 Results

Table 2 summarizes the matched comparison. The ODE predictor improves fit/loss (0.234 vs. 0.248) and val/loss (0.506 vs. 0.536); final val prediction / SIGReg terms are approximately 0.220 / 3.17. CEM success is identical: **5 successes out of 93** feasible val starts ($\approx 5.38\%$). ODE CEM wall-clock is substantially higher (≈ 82 min vs. ≈ 8 min for discrete on the same machine) because each rollout step runs four RK4 stages.

Protocol note. Earlier CEM probes on random full-dataset starts (e.g. 0/10) are *not* val-tied and are discarded; all claims here use the val-99 / 93-feasible protocol shared with `validate/loss`.

4.3 Limitations

- **1k subset.** Results are on 1000 windows, not the full PushT expert corpus. Paper-scale $\approx 96\%$ success is out of scope for this study.

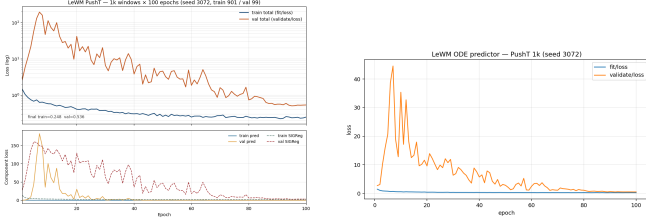


Figure 1. Training curves on the fixed 1k split (left: discrete AR; right: ODE predictor). **ODE lower loss does not yield higher CEM success under this budget; planning may be bottlenecked by data scale, CEM hyperparameters, or goal-offset feasibility rather than predictor form.**

- **Single environment.** Only PushT is evaluated; Cube, Two-rooms, and Reacher remain future work.
- **Fixed solver.** We use RK4 with $N=4$ and no adaptive stepping; broader ODE solvers and multi-step τ schedules are unexplored.

5 Conclusion

We presented a minimal continuous-time drop-in for LeWM-style JEPA world models: an action-conditioned ODE predictor whose one-step map is RK4 integration from $\tau=0$ to $\tau=1$. On official PushT data with a fixed 1k-window split, the ODE model matches discrete AR CEM success (5/93) while reducing train and validation loss. The result supports continuous latent dynamics as a viable predictor inductive bias for planning JEPA models, but—honestly—does not yet improve CEM under the 1k regime, nor close the gap to paper-scale PushT performance. Natural next steps are full-dataset training, multi-environment CEM, and stronger continuous-time formulations in the spirit of ODEWorld [8].

References

- [1] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *CVPR*, 2023.
- [2] R. Balestriero et al. The birth of SIGReg: Sketched isotropic Gaussian regularization. *arXiv preprint*, 2023. As used in LeWM; see Maes et al. 2026.
- [3] A. Bardes, J. Ponce, and Y. LeCun. V-JEPA: Latent video prediction for self-supervised learning. *arXiv preprint arXiv:2404.08471*, 2024.
- [4] R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud. Neural ordinary differential equations. In *NeurIPS*, 2018.
- [5] K. Chua, R. Calandra, R. McAllister, and S. Levine. Deep reinforcement learning in a handful of trials using probabilistic dynamics models. In *NeurIPS*, 2018.
- [6] D. Ha and J. Schmidhuber. World models. In *NeurIPS*, 2018.
- [7] D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson. Learning latent dynamics for planning from pixels. In *ICML*, 2019.
- [8] D. Liu et al. ODEWorld: A continuous predictive architecture via physical-time flow. *arXiv preprint arXiv:2607.27924*, 2026.
- [9] L. Maes, Q. Le Lidec, D. Scieur, Y. LeCun, and R. Balestriero. LeWorld-Model: Stable end-to-end joint-embedding predictive architecture from pixels. *arXiv preprint arXiv:2603.19312*, 2026.